

HEALTHCARE DATA CLIENT

Exception Analysis

Healthcare Data Quality Automation and Exception Reporting

Document Control	Detail
Engagement	Jul-Oct 2026
Role	Data Analyst
Primary Stakeholder	Healthcare Analyst
Batch Scale	Approximately 250,000 records per batch
Observed Exception Rate	2%
Review Turnaround	Reduced from 6 hours to 2.5 hours per file
Analysis boundary: The 2% exception rate is confirmed. Exact historical counts by validation rule, healthcare domain, source, reason code or batch are not retained and are therefore not fabricated in this analysis.	

1. Analysis Objective

The exception analysis focuses on records that fail one or more applicable data-quality rules. Its purpose is to separate records requiring attention from the processed population, preserve the reason for each failure and support targeted Healthcare Analyst review.

2. Exception Population

Measure	Definition	Confirmed / Derived Result
Records Processed	Total governed records included in a batch.	Approximately 250,000 per batch
Exception Records	Distinct processed records failing one or more applicable validation rules.	Exact count not retained
Exception Rate	Exception Records / Records Processed x 100.	2% observed
Validation Failures	Total rule-level failures across records.	Exact count not retained
Review Turnaround	Elapsed review time per file.	6 hours reduced to 2.5 hours

Counting control: A record failing multiple rules is counted once as an Exception Record, but may contribute multiple Validation Failures.

3. Exception Categories

Exception Category	Confirmed Analytical Meaning	Historical Count
Duplicate	Record met the implemented duplicate condition.	Not retained
Missing Value	Applicable value was null, blank or missing.	Not retained
Invalid Date	Applicable date failed the implemented date-validity rule.	Not retained
Incorrect Format	Value failed the documented expected-format rule.	Not retained
Invalid ID	Identifier failed the documented identifier rule.	Not retained
Out-of-Range Value	Value fell outside documented valid bounds.	Not retained
Inconsistent Code	Value did not satisfy the documented code set or mapping.	Not retained
Cross-Field Mismatch	Related fields did not satisfy the documented relationship.	Not retained

4. Exception Analysis Framework

Analysis Lens	Question Answered	Evidence Requirement
Reason code	Why was the record flagged?	Rule-level reason or failure description
Validation rule	Which check failed?	Rule ID or governed rule name
Source / batch	Where did the exception originate?	Source or batch reference
Healthcare domain	Which confirmed data domain is associated with the record?	Source-supported domain classification
Multi-failure records	Did one record fail multiple rules?	Rule-level failure grain
Record-level status	Does the record require review?	Exception flag
Review output	Can the analyst investigate the record?	Traceable exception file, table or report

5. Reason-Code Analysis

Reason codes convert technical validation failures into reviewable exception categories. They allow the analyst to understand why a record was flagged without manually rediscovering the failure condition.

Control	Requirement
Reason assignment	Each failed rule produces an identifiable reason code or equivalent failure explanation.
Consistency	The same rule uses consistent reason classification across recurring batches.
Multiple reasons	A record can retain more than one reason when multiple rules fail.
Traceability	Reason information remains linked to the affected source record.
Reporting	Reason-level analysis uses rule-level detail rather than duplicating record-level Exception Rate.
Historical boundary	Exact original reason-code labels and frequencies are not retained.

6. Multi-Failure Analysis

Scenario	Record-Level Treatment	Rule-Level Treatment
Record passes all applicable rules	Non-exception record	No validation-failure row
Record fails one rule	One Exception Record	One validation-failure row
Record fails multiple rules	Still one Exception Record	Multiple validation-failure rows
Exception Rate	Count the affected record once	Do not calculate record-level rate from failure-row count

Why this matters: Separating record grain from failure grain prevents the 2% Exception Rate from being inflated by records that fail more than one rule.

7. Healthcare Domain Analysis

Confirmed Domain	Permitted Portfolio Analysis
Claims	Analyze exceptions when source records are explicitly identified as claims.
Patients	Analyze exceptions when source records are explicitly identified as patient data.
Appointments	Analyze exceptions when source records are explicitly identified as appointments.
Providers	Analyze exceptions when source records are explicitly identified as provider data.
Billing	Analyze exceptions when source records are explicitly identified as billing data.
Transactions	Analyze exceptions when source records are explicitly identified as transactions.
Clinical records	Analyze exceptions when source records are explicitly identified as clinical records.
Insurance data	Analyze exceptions when source records are explicitly identified as insurance data.

No fabricated distribution: The portfolio must not claim that one healthcare domain had more exceptions than another because the historical domain-level exception distribution is not retained.

8. Review Prioritization

The confirmed project evidence supports targeted exception review through separate outputs, highlighting and reason codes. It does not support an exact historical severity model, priority score or clinical-risk ranking. Any future prioritization framework must therefore be presented as a proposed design rather than as historical client logic.

9. Exception Output Requirements

Output Element	Purpose
Source record reference	Reconcile the exception to the processed source record.
Exception flag	Identify that the record requires review.
Reason code / failure explanation	Explain why the record was flagged.
Validation rule	Identify the check responsible for the failure.
Batch reference	Connect the exception to its processing context.
Relevant review context	Display only source-supported fields needed for controlled investigation.

10. Reconciliation & Quality Controls

Control	Expected Result
Population control	Exception Records are a subset of Records Processed.
Failure control	Every Exception Record has at least one validation failure.
Reason control	Each validation failure has an identifiable reason or rule.
Double-counting control	Multi-failure records count once in record-level Exception Rate.
Rate control	Exception Rate reconciles to Exception Records / Records Processed x 100.
Observed result control	The retained overall Exception Rate is 2%.
Reporting control	Power BI exception measures reconcile to governed validation output.

11. Confirmed Analytical Findings

Finding	Evidence-Safe Interpretation
2% exception rate	A small but material share of processed records required review under the implemented validation logic.
Exception outputs generated	Failed records were separated into a file, table or report for targeted review.
Reason-coded review	Exceptions were highlighted and assigned a reason code or equivalent explanation.
High-volume processing	The validation process operated across approximately 250,000 records per batch.
Turnaround improvement	Review turnaround reduced from 6 hours to 2.5 hours per file.

Interpretation boundary: The evidence does not establish which exception category was most common, which healthcare domain had the highest exception rate, or whether the exceptions affected clinical outcomes.

12. Business Impact

Outcome	Result
Baseline review turnaround	6 hours per file
Post-solution review turnaround	2.5 hours per file
Reduction	3.5 hours per file
Derived reduction percentage	Approximately 58.3%
Interpretation	Reduced review turnaround through automated validation and targeted exception review

13. Traceability

Analysis Area	Upstream Artifact	Downstream Artifact
Exception definition	BRD; KPI Catalogue	Power BI Requirements; Testing
Validation categories	Data Quality Rule Catalogue	Exception reporting; Testing
Reason-code analysis	Data Quality Rule Catalogue	Dashboard; Findings
Record vs failure grain	Data Dictionary; KPI Catalogue	Data model; Dashboard
2% Exception Rate	KPI Catalogue	Findings; Final Report
Review turnaround	KPI Catalogue	Findings; Final Report